

```
/usr/bin/python3.8 /home/student/Sreehari C/Sklearn/exp31.py
```

First few rows of the dataset:

|   | Id | SepalLengthCm | SepalWidthCm | PetalLengthCm | PetalWidthCm | Species     |
|---|----|---------------|--------------|---------------|--------------|-------------|
| 0 | 1  | 5.1           | 3.5          | 1.4           | 0.2          | Iris-setosa |
| 1 | 2  | 4.9           | 3.0          | 1.4           | 0.2          | Iris-setosa |
| 2 | 3  | 4.7           | 3.2          | 1.3           | 0.2          | Iris-setosa |
| 3 | 4  | 4.6           | 3.1          | 1.5           | 0.2          | Iris-setosa |
| 4 | 5  | 5.0           | 3.6          | 1.4           | 0.2          | Iris-setosa |

Feature data (first 5 rows):

|   | Id | SepalLengthCm | SepalWidthCm | PetalLengthCm |
|---|----|---------------|--------------|---------------|
| 0 | 1  | 5.1           | 3.5          | 1.4           |
| 1 | 2  | 4.9           | 3.0          | 1.4           |
| 2 | 3  | 4.7           | 3.2          | 1.3           |
| 3 | 4  | 4.6           | 3.1          | 1.5           |
| 4 | 5  | 5.0           | 3.6          | 1.4           |

Labels (first 5 rows):

|   |             |
|---|-------------|
| 0 | Iris-setosa |
| 1 | Iris-setosa |
| 2 | Iris-setosa |
| 3 | Iris-setosa |
| 4 | Iris-setosa |

Name: Species, dtype: object

Training features (first 5 rows):

|    | Id | SepalLengthCm | SepalWidthCm | PetalLengthCm |
|----|----|---------------|--------------|---------------|
| 22 | 23 | 4.6           | 3.6          | 1.0           |
| 15 | 16 | 5.7           | 4.4          | 1.5           |
| 65 | 66 | 6.7           | 3.1          | 4.4           |
| 11 | 12 | 4.8           | 3.4          | 1.6           |
| 42 | 43 | 4.4           | 3.2          | 1.3           |

Testing features (first 5 rows):

|     | Id  | SepalLengthCm | SepalWidthCm | PetalLengthCm |
|-----|-----|---------------|--------------|---------------|
| 73  | 74  | 6.1           | 2.8          | 4.7           |
| 18  | 19  | 5.7           | 3.8          | 1.7           |
| 118 | 119 | 7.7           | 2.6          | 6.9           |
| 78  | 79  | 6.0           | 2.9          | 4.5           |
| 76  | 77  | 6.8           | 2.8          | 4.8           |

GaussianNB()

```
Array: ['Iris-versicolor' 'Iris-setosa' 'Iris-virginica' 'Iris-versicolor'
'Iris-versicolor' 'Iris-setosa' 'Iris-versicolor' 'Iris-virginica'
'Iris-versicolor' 'Iris-versicolor' 'Iris-virginica' 'Iris-setosa'
'Iris-setosa' 'Iris-setosa' 'Iris-setosa' 'Iris-versicolor'
'Iris-virginica' 'Iris-versicolor' 'Iris-versicolor' 'Iris-virginica'
'Iris-setosa' 'Iris-virginica' 'Iris-setosa' 'Iris-virginica'
'Iris-virginica' 'Iris-virginica' 'Iris-virginica' 'Iris-virginica'
'Iris-setosa' 'Iris-setosa']
```

Actual labels: 73 Iris-versicolor

```
18      Iris-setosa
118     Iris-virginica
78      Iris-versicolor
76      Iris-versicolor
31      Iris-setosa
64      Iris-versicolor
141     Iris-virginica
68      Iris-versicolor
82      Iris-versicolor
110     Iris-virginica
12      Iris-setosa
36      Iris-setosa
9       Iris-setosa
19      Iris-setosa
56      Iris-versicolor
104     Iris-virginica
69      Iris-versicolor
55      Iris-versicolor
132     Iris-virginica
29      Iris-setosa
127     Iris-virginica
26      Iris-setosa
128     Iris-virginica
131     Iris-virginica
145     Iris-virginica
108     Iris-virginica
143     Iris-virginica
45      Iris-setosa
30      Iris-setosa
Name: Species, dtype: object
```

```
Confusion Matrix: [[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
```

Accuracy: 1.0

Process finished with exit code 0